Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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Abstract

Linear probes tell us that emotion becomes *readable* in the middle-to-late layers of frozen speech transformers — but not *which* internal features carry it, and not whether a probe is listening to *how* something is said (acoustics) or *what* is said (words). We open these models up with Sparse Autoencoders (SAEs): we train a TopK SAE on the per-frame activations of six frozen large encoders (HuBERT, WavLM, wav2vec2, w2v-BERT, MERT, Whisper) and read off the individual features it learns. The trick that makes this work is a lexicon-controlled corpus, CREMA-D (91 actors × 12 *fixed* sentences × 6 emotions), plus a confound check that keeps a feature only if it tells us more about emotion than about the sentence or the speaker — otherwise the SAE just rediscovers phonemes. Four findings follow: (i) emotion is *organised* into clean features earlier (layers 2–7) than where a probe reads it best (layer 10); (ii) all six encoders share a common, speaker-invariant acoustic feature basis; (iii) the recovered signal is acoustic, not lexical; and (iv) on incongruent speech (EMIS), the "lexical shortcut" is a *deep-layer* habit that only the ASR-trained encoders pick up, while Whisper stays grounded in the voice at every layer.

Keywords

speech emotion recognition, sparse autoencoders, interpretability, disentanglement, acoustic features

1 Why this study

Self-supervised speech encoders sit on top of the speech-emotion leaderboards, but they were originally trained to transcribe words, so they carry a lot of linguistic information. That raises an uncomfortable question: when a probe reads emotion out of one of these models, is it hearing the *tone of voice*, or is it quietly reading the *words*? A model that leans on the words will fail exactly where we most need it — sarcasm, accents, and languages where the lexicon is unfamiliar.

Standard probing cannot answer this. It tells us *where* emotion is decodable, but treats the layer as a black box. We instead use a Sparse Autoencoder, which factors the dense activation of a layer into a large dictionary of sparse, mostly single-meaning "features." Inspecting those features lets us ask *which* ones track emotion, *what* they respond to (prosody vs. words), and *whether* they are shared across models and speakers. In short, we move from **decodability** to **mechanism**.

A known hazard is that an SAE trained on ordinary speech mostly rediscovers *phonemes* [4, 9]. We sidestep it by anchoring on CREMA-D, where the 12 carrier sentences are identical across emotions, so a recovered emotion signal cannot be coming from the words. This doubles as a built-in sanity check.

What we contribute. (1) a confound-controlled recipe for finding genuine emotion features in speech SAEs; (2) the observation that disentanglement and decodability peak at *different* depths; (3) an acoustic-vs-lexical label per feature plus a causal sufficiency test; and (4) a depth-resolved picture of the lexical shortcut on EMIS. Code and artifacts are released.

2 How we did it

Activations. The project's cached embeddings are averaged over time (one vector per clip), which is far too few examples to fit a large SAE. So we re-extract the *un-pooled, per-frame* hidden states of all six frozen encoders — 25 layers each (a CNN front-end plus 24 transformer layers), with Whisper's padding frames masked out. CREMA-D layer 13 alone gives 940,273 frames to train on.

The SAE. A TopK SAE (d_{in}=1024, dictionary size 8192, k=32 active features per frame), with a unit-norm decoder and a small auxiliary loss that keeps features from dying. It is a faithful compressor: it explains **93%** of the activation variance with essentially no dead features (Table 3).

Finding the emotion features — and the confound check. For every feature we measure how much its on/off firing tells us about the emotion label (mutual information, with a shuffle test for significance). The catch: at this scale almost everything is "significant." What separates a real emotion feature from a phoneme detector is (a) *monosemanticity* — most of its activity falls on one emotion — and (b) the *confound check*: it must tell us more about emotion than about the spoken sentence (a stand-in for phonetics) *or* the speaker. Significance plus monosemanticity alone flags 394 features, but two-thirds of those are really about the sentence or the speaker. After the confound check, **61** clean emotion features survive (Table 3); they are dominated by anger and sadness, while fear and neutral drop to zero — exactly the high-arousal emotions that have the clearest acoustic signature.

Acoustic or lexical? For each emotion feature we ask whether its firing is better predicted by acoustic descriptors (eGeMAPS [6]) or by the sentence identity (the lexical content). We use classification AUC rather than R^2 because the features are about 96% silent.

3 What we found

3.1 Emotion is organised earlier than it is decoded

If you just train a probe, accuracy climbs and peaks *deep* (layer 10) and stays high. But the *clean* emotion features tell a different story: their count peaks *early* (layer 7), and their purity even earlier (layer 2). By the deep layers, emotion is still easy to decode but has smeared out across many directions — only 24–47 clean features remain (Fig. 1, top-left; Fig. 2, left). In plain terms: **the model tidies emotion into neat features early, then a probe reads it best later, once it has been re-mixed for linear separability.** Decodability and disentanglement are not the same thing.

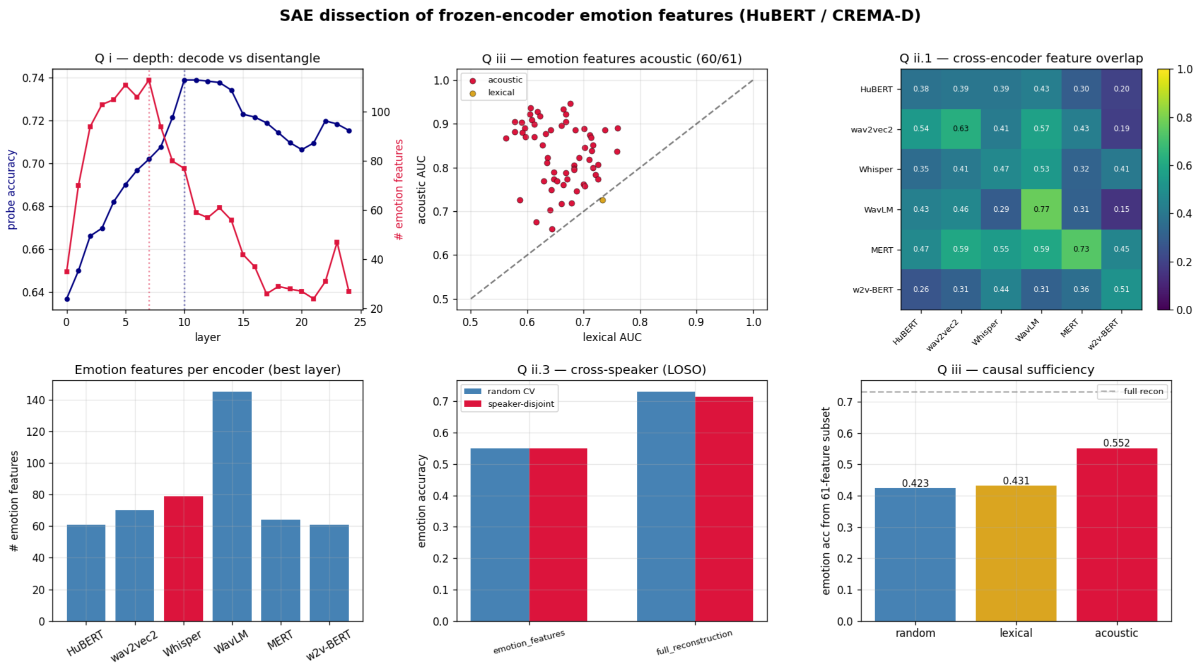

Figure 1: The whole study at a glance (HuBERT / CREMA-D). Top: depth — probe accuracy vs. number of clean emotion features per layer; each feature acoustic vs. lexical (above the line = acoustic); how features overlap across the six encoders. Bottom: number of emotion features per encoder; cross-speaker test (random vs. leave-speaker-out); and the causal test (how well emotion decodes from only-acoustic vs. only-lexical vs. random features).

3.2 The same acoustic features, across encoders and speakers

Every encoder ends up with a healthy set of emotion features (Table 1), all anger/sadness-dominated. When we match features across encoders by how they fire, Whisper overlaps with the self-supervised models just as much as they overlap with each other (0.40 vs. 0.39). So on fixed-lexicon speech, *Whisper is not special* — its much-discussed difference is about text, which CREMA-D cannot expose (we return to it in §3.4). The features also travel across people: under a strict leave-one-speaker-out test over all 91 actors, accuracy barely moves (drop ≈ 0) and each feature fires for ∼95% of the actors who express its emotion. This also explains a known result elsewhere: the large speaker-out drop on MESD (∼0.30) comes from *speaker-entangled* features — precisely the ones our confound check throws away.

3.3 The signal is acoustic, not lexical

On fixed-lexicon CREMA-D, an emotion feature *must* be acoustic — and 60 of 61 are (mean acoustic AUC 0.83 vs. 0.66 for sentence identity; Fig. 1, top-middle). The causal test agrees: if we let emotion be decoded only from the acoustic features we get 0.55, but only from the lexical features just 0.43 — barely above the 0.42 we get from random features (Fig. 1, bottom-right). The acoustic features are *sufficient*; the lexical ones are not.

Table 1: Per-encoder emotion features at each encoder's best CREMA-D layer.

Encoder	Best layer	# emotion feats	Family
HuBERT	13	61	SSL (ASR-style)
wav2vec2	10	70	SSL (ASR-style)
Whisper	12	79	Supervised ASR
WavLM	14	145	SSL (ASR-style)
MERT	9	64	Music SSL
w2v-BERT	17	61	SSL (multilingual)

3.4 The lexical shortcut is a deep-layer habit

CREMA-D cannot test the text shortcut, because its text is fixed. EMIS [5] can: it is synthetic speech where the *written* emotion and the *spoken* emotion are deliberately mismatched. We train a probe on the matched clips and test on the mismatched ones; "text-bias" is how often the model goes with the words instead of the voice. The result (Table 2, Fig. 3): every encoder is grounded in the voice early on, but the three ASR-trained self-supervised models (HuBERT, WavLM, wav2vec2) *develop* the shortcut in their deep layers. Whisper never does — it stays voice-grounded at every layer — and the music model MERT and w2v-BERT never take it either. This reproduces the previously published per-encoder text-bias and adds the new part of the story: the shortcut lives at a depth Whisper simply never reaches.

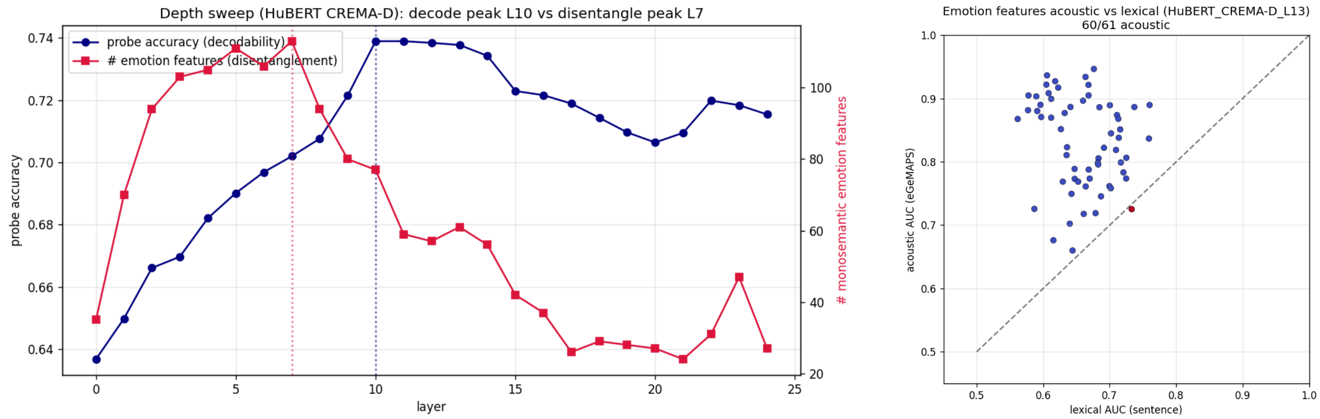

Figure 2: Left: depth — a probe reads emotion best deep (L10) while the count of clean emotion features peaks early (L7). Right: each emotion feature scored on acoustic (eGeMAPS) vs. lexical (sentence) prediction — nearly all sit on the acoustic side of the diagonal.

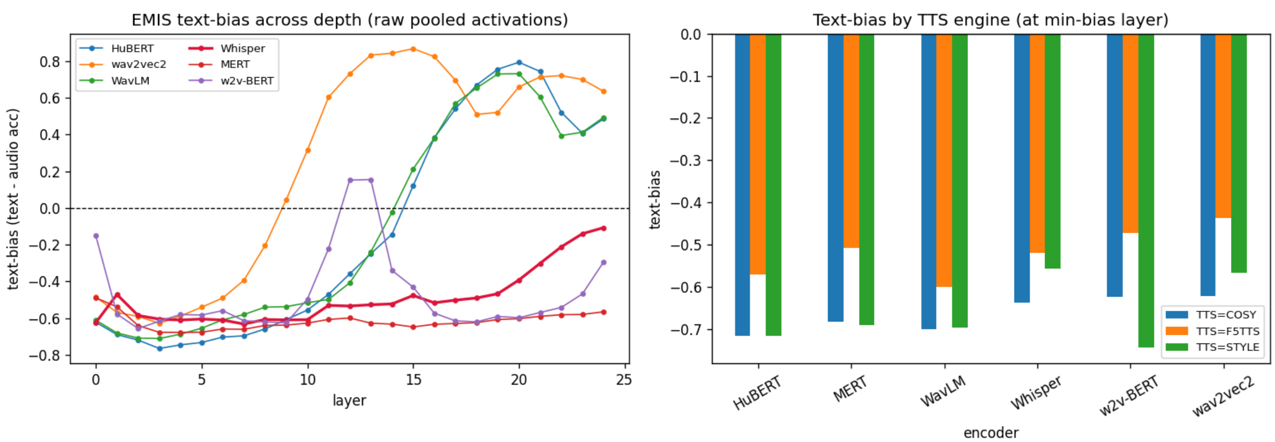

Figure 3: EMIS text-bias (text − audio accuracy) across all 25 layers. The "use-the-words" shortcut appears only in the *deep* layers of the ASR-trained encoders (HuBERT/WavLM/wav2vec2); Whisper (red) stays voice-grounded at every layer, and MERT/w2v-BERT never take it. Right panel: the same broken down by TTS engine.

Table 2: EMIS text-bias (text − audio accuracy) at each encoder's most text-biased layer; positive = takes the lexical shortcut. Raw mean-pooled activations.

Encoder	Text-bias	Verdict
wav2vec2 (L15)	+0.87	takes the shortcut
HuBERT (L20)	+0.79	takes the shortcut
WavLM (L20)	+0.73	takes the shortcut
Whisper (L24)	**−0.11**	**resists (voice-grounded)**
MERT	−0.60	resists (music model)
w2v-BERT	−0.60	resists

Table 3: Key numbers (HuBERT / CREMA-D, layer 13 unless noted).

Quantity	Value
SAE reconstruction (variance explained)	93% (FVU 0.07)
Dictionary / sparsity / dead features	8192 / k=32 / 4
Emotion feats: significant → clean → confound-OK	6407 → 394 → 61
Decode peak / disentangle peak / purity peak	L10 / L7 / L2
Acoustic features (positive control)	60/61
Decode from acoustic / lexical / random feats	0.55/0.43/0.42
Cross-speaker leave-one-out drop	≈ 0
EMIS: Whisper L24 / mean ASR-SSL text-bias	−0.11/+0.80

4 The pipeline, and what is released

The study runs as nine numbered, reproducible steps (Table 4), each a module under `src/sae/steps/` callable as `python -m src.sae.steps`

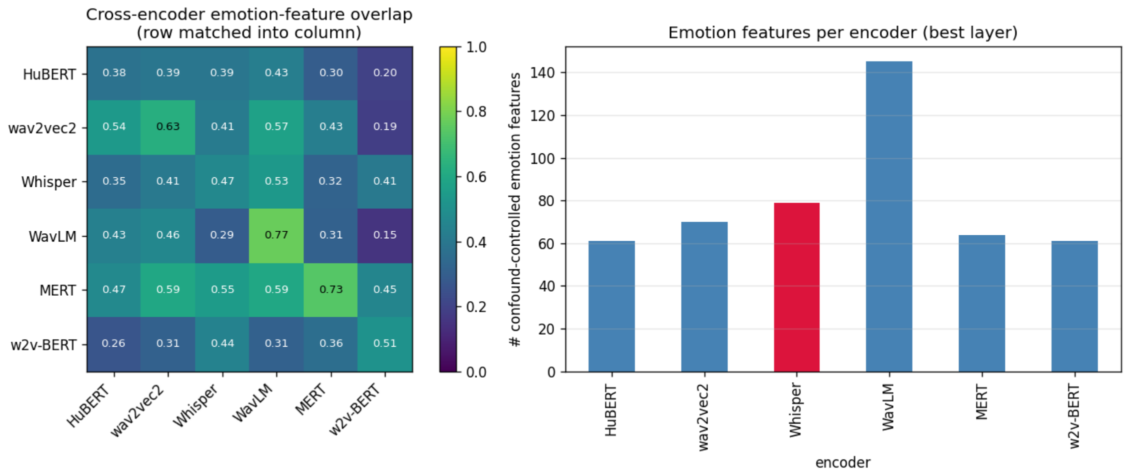

Figure 4: Cross-encoder picture. Left: how much each encoder's emotion features overlap with every other's — Whisper is not an outlier. Right: number of clean emotion features per encoder at its best layer.

Table 4: The pipeline. Each step is one reproducible command.

Step	Does	Headline output
0	inventory the cached embeddings	coverage map
0.5	extract per-frame activations	940k frames
1	train a TopK SAE	93% variance, 4 dead
2	find emotion features (confound-ctrl)	394 → 61
3	label acoustic vs. lexical	60/61 acoustic
4	depth sweep (all 25 layers)	L10 vs. L7
5	cross-encoder overlap	Whisper not special
6	cross-speaker (leave-one-out)	drop ≈ 0
7	causal sufficiency	0.55 vs. 0.43
8	EMIS text-bias by depth	shortcut is deep

`<step>`. Heavy jobs (extraction, the 25-layer SAE grid) fan out as SLURM array jobs on a single V100 each. Everything needed to reproduce a number is versioned: the per-step CSVs, plots, and metrics live under `results/SAE/`; the large activation tensors and SAE checkpoints live on scratch (too big for version control) with pointers recorded. CREMA-D was pulled from a public mirror and EMIS from its open Zenodo release.

5 Related work and how we differ

The closest method paper is *AudioSAE* [3] (EACL 2026): the same SAE recipe (TopK, 8×, per-frame, all layers) on two encoders, even with a cross-model feature-overlap analysis. But it treats emotion as just one of four generic probe tasks — no dedicated emotion-feature interpretation, no confound check, no lexicon-controlled corpus, and no acoustic-vs-lexical split. Two further SAE-for-speech papers find phonetic features [9] or singing-technique features [8], not affect, and do not sweep depth. The closest mechanistic SER paper [7] uses probing, logit-lens and CKA but *no* SAEs; the acoustic-vs-lexical question itself is studied with text-dependency analyses [1, 2] and the EMIS benchmark [5]. Our specific combination — confound-controlled emotion features on a lexicon-controlled factorial corpus across six encoders, plus the depth and EMIS results — is, to our knowledge, new.

6 Limitations and next steps

Because emotion is spread across thousands of features, removing a handful barely dents accuracy, so we rely on the *sufficiency* test rather than ablation. One caution we learned the hard way: the SAE's *reconstruction* must not stand in for raw activations in the EMIS probe — doing so spuriously flips Whisper. Most depth results are on HuBERT/CREMA-D; the cross-encoder and EMIS arms use one layer per encoder rather than full 25-layer grids. Two follow-ups are coded and waiting on data: a causal ablation on IEMOCAP with *real* transcripts, and a SpeechCraft cross-lingual study to explain the published English↔Chinese collapse as disjoint feature sets. Beyond those: feature *steering* (turning an emotion feature up to shift predictions) would upgrade our correlational claims to causal ones.

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